I Overview

A) Introduce one of the simplest ways to overcome disturbances – Integrals

1) Set point control where set point is previously unknown or may change

2) Used for Overcoming noise

This lecture does <u>not</u> depend on LQR, you could also use pole placement.

II Recall Simple Recipe for State Feedback Control

A) Step 0: Determine the linear system (A, B) to be controlled

1) Not always trivial $\dot{x} = Ax + Bu$

2) Science: Various types of linearization x_d: desired state

3) Art: What states to control

B) Change of State

1) Make desired to be zero-state

$$z = x - x_d$$

C) Find Feed-Forward Term

1) Cancel out terms you don't want

$$Bu_{ff} + Ax_d - \dot{x}_d = 0$$

D) Express system using full control

1) Sanity check

$$u = \hat{u} + u_{ff} \rightarrow \dot{z} = Az + B\hat{u}$$

E) Check Controllability

1) Can you place eigenvalues where you want?

$$\text{rank}(\mathcal{M}) = n$$

F) Use "pole placement" to choose eigenvalues

$\hat{u} = -Kz \quad \dot{z} = \bar{A}z$

G) Express full control

1) Sanity check

$u = u_{ff} - Kz = u_{ff} - K(x - x_d)$

III Reference Tracking Example $\ddot{q} + \dot{q} + 4q = 10u$

$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix} \quad \dot{x} = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 10 \end{bmatrix} u \qquad q_d = r \quad \dot{q}_d = \dot{r}$

$x_d = \begin{bmatrix} r \\ \dot{r} \end{bmatrix}$

$z = x - x_d$

$\Rightarrow \quad \dot{z} = Az + Bu \Rightarrow \dot{x}_d = Ax_d + Bu_{ff}$

$\Rightarrow \quad u_{ff} = \frac{4r + \dot{r} + \ddot{r}}{10}$

$\Rightarrow \quad K = lqr(A, B, Q, R)$

↳ Choose Q & R

$R = 1 \qquad Q = \begin{bmatrix} q_1 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow G = [1 \;\; 0]z$

$Q = q_1$

$\hat{Q} = G^T Q G$

Does it work? $R > 0 \quad Q \geq 0$

Check Observability

$\sqrt{Q} = [\sqrt{q_1} \;\; 0] = C$

$\Rightarrow C^T C = Q$

check observability of (A, C_c)

$$\text{rank}(\mathcal{O}) = 2$$

$$u = \frac{4r + \dot{r} + \ddot{r}}{10} - k(x - x_d)$$

IV The solution we found above works great, but when we introduce noise, the system is offset.

$$\dot{x} = Ax + Bu + \underbrace{Ew}_{\text{Noise}}$$ — Assuming constant noise for now.

V Disturbance Rejection

A) Idea: Use Integral to help overcome disturbance.

↳ For state space Integral design that means to just introduce a new state.

1) We have

$$\left.\dot{z} = Az + B\hat{u}\right\}$$ Our designed control

$\dot{\sigma}_1$ σ_1 let $\sigma_1 = Cz$

$(e, \int e\,dt)$

↳ Can have multiple rows if you care about multiple things.

In the case of our example $C = \underline{[1 \;\; 0]}$

2) Form augmented state vector

$$\xi = \begin{bmatrix} z \\ \sigma \end{bmatrix} \qquad \dot{\xi} = \underbrace{\begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix}}_{\hat{A}} \begin{bmatrix} z \\ \sigma \end{bmatrix} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{\hat{B}} \hat{u}$$

$\text{rank}(\mathcal{M}) = 3$ for $\mathcal{M}(\hat{A}, \hat{B})$

check controllability

Use Augmented State

Now that we have a new $\hat{A}$ & $\hat{B}$ we can say

$$\hat{u} = -K\xi \qquad Q = \begin{bmatrix} q_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & q_3 \end{bmatrix}$$

We can weight our integral

$$R = 1$$

$\underline{Q \succeq 0 \quad R \succ 0}$

Check Observability

$$C_c = \begin{bmatrix} \sqrt{q_1} & 0 & 0 \\ 0 & 0 & \sqrt{q_3} \end{bmatrix}$$

$$\Rightarrow C_c^T C_c = Q$$

$$\text{Rank}(\text{Obsv}(\hat{A}, C_c)) = 3 \Rightarrow \text{C.O.}$$

B) Ramp Disturbance

For our previous system, we have a similar problem as the constant noise problem.

1) Use a double integrator.

Introduce a new state $\left(\overset{\dot{\sigma}_1}{e}, \overset{\sigma_1}{\int e\,dt}, \overset{\sigma_2}{\iint e\,dt}\right)$

$$\xi = \begin{bmatrix} z \\ \sigma_1 \\ \sigma_2 \end{bmatrix} \qquad \Rightarrow \underline{\dot{\sigma}_2 = \sigma_1}$$

$$\dot{\xi} = \underbrace{\begin{bmatrix} A & 0 & 0 \\ C & 0 & 0 \\ 0 & I & 0 \end{bmatrix}}_{\hat{\hat{A}}} \begin{bmatrix} z \\ \sigma_1 \\ \sigma_2 \end{bmatrix} + \underbrace{\begin{bmatrix} B \\ 0 \\ 0 \end{bmatrix}}_{\hat{\hat{B}}} u$$

2) Going back to the example

$$K = lqr(\hat{A}, \hat{B}, Q, R)$$

$$Q = \begin{bmatrix} q_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & q_3 & 0 \\ 0 & 0 & 0 & q_3 \end{bmatrix} \qquad Q = C_c^T C_c$$

$$\text{rank}(\text{obsv}(\hat{A}, C_c)) = 4 \Rightarrow C_1 C_1$$

c) Even more complicated Disturbance

$$\dot{\sigma}_3 = \sigma_2$$

$$\xi = \begin{bmatrix} z_1 \\ \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix}$$

$$\dot{\xi} = \begin{bmatrix} A & 0 & 0 & 0 \\ C & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix} \begin{bmatrix} z \\ \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} +$$

↓

Check controllability, design Q, R, create control

1) The error becomes better, but it does not go away entirely

D) Use knowledge of noise for better results (Not on final)

1) Assume LTI dynamics with disturbance

$$\dot{x} = Ax + Bu + Ew$$

$$y = Cx + Du$$

$$w^{(p)} = \sum_{i=1}^{p} a_i \, w^{(p-i)} \; ; \quad w(0) : \text{unknown}$$

Use to define "noise characteristics"